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Urbanization and Energy Equity: An Urban-rural Gap Perspective

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Abstract

Inequality is usually discussed in a monetary perspective, while the disparity in energy access sometimes is more severe. This paper mainly studies the impact of urbanization on urban-rural electricity consumption inequality with a modified STRIPAT model. We find that there is a long-run U-shaped relationship between urbanization and urban-rural electricity consumption inequality, and the estimated turning point arrives at the urbanization level of around 60%. We further carry out a regional heterogeneity analysis and then have two interesting findings: firstly, the colder northern region's turning point arrives later than the south. Secondly, the baseline U-shaped relationship remains for developed eastern regions, while for the undeveloped midwest, the relationship is not nonlinear but linearly negative. As an extension, we lastly explore the mechanism underlying the U-shaped relationship, and find that the interaction of urbanization's scale and efficiency effect determines the U-shaped relationship.

Keywords: Urbanization, Urban-rural electricity consumption inequality,
STRIPAT, U-shaped relationship, Scale effect, Efficiency effect

1 Introduction

Inequality is usually discussed and analyzed in a monetary perspective. However, a growing literature has shown that disparities in energy access and use are sometimes starker than inequalities in income¹ (Wu and Li, 2008; Pachauri and Rao, 2013; Sovacool and Dworkin, 2015; Wu et al, 2017). In particular, it seems to be more severe in electricity consumption disparity: worldwide, there are 840 million population without electricity access in 2017 (World Bank ESMAP, 2019). For China, its electricity consumption inequality is also serious. For example, in 2016, the traditional biofuels consumption² of China's rural habitants with no electricity access is equivalent to 1.3 million tons of coal, and the indoor pollution arising from it leads to around 0.8 to 1.5 million premature deaths per year (WHO, 2016).

Meanwhile, China is experiencing a rapid urbanization, which is only 31.91% in 1997 while 57.35% in 2016. Such high-speed urbanization has a substantial impact on citizens' electricity consumption by raising their purchasing power, changing their energy behavior and improving power facilities (Rokeach, 1973; Karp, 1996; Stern, 2000). The below Figure 1 may be informative: in 1997, the urbanization rates of all China's provinces are relatively low and corresponding gaps between urban and rural electricity consumption per capita are quite large. In 2016, most provinces' urbanization increases a lot and corresponding urban-rural electricity consumption gaps experience a tremendous drop. Even surprisingly, some developed eastern provinces' rural electricity consumption has exceeded the urban.

That phenomenon raises some questions that whether there is a robust relationship between urbanization and urban-rural electricity consumption inequality and how urban-rural electricity consumption inequality changes in response to urbanization. The major task of this paper is to find the answer to this unexplored question. The answer will help China's policymakers find ways to ensure people's access to adequate, reliable and affordable supplies of electricity, which further relates to the promotion of productive opportunities, enterprise growth and employment, and has a positive impact on particularly the health of rural women and children who are often exposed to harmful emissions associated with inefficient cooking using solid fuels.

We start our analysis by building a modified STRIPAT model. To capture urban-rural electricity consumption inequality in the model, we introduce the Theil index. Besides, we also take the simple way: taking the ratio

¹The poorest 40% of world's population consumes only about 10% of world's final energy production while the richest 30% disposes about 60% (Pachauri and Rao, 2013).

²Biofuels, any fuel that is derived from biomass—that is, plant or algae material or animal waste. The most common biofuels in China's rural are firewoods and animal waste.

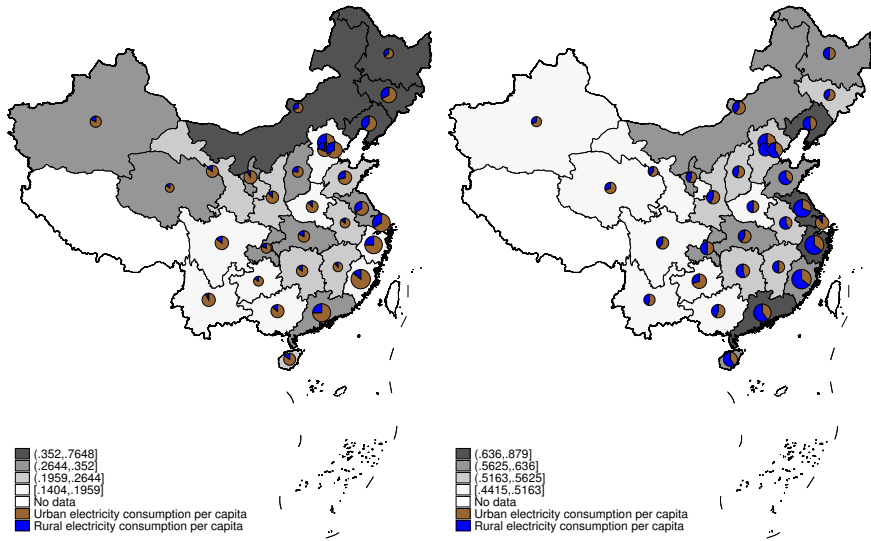


Fig. 1 Urbanization and urban-rural residential electricity consumption in 1997 and 2016

of urban electricity consumption per capita to the rural as another measurement of urban-rural electricity consumption inequality in the following empirical analysis. Then the model decomposes inequality determinants into urbanization, urban-rural income disparity and other socio-economic factors. Following [Martínez-Zarzoso and Maruotti \(2011\)](#), to investigate the potential Environmental Kuznets Curve (EKC) relationship between urbanization and environmental impacts, we add the quadratic term of urbanization into the model. The data we used is a provincial panel data including 30 China's provinces over the period of 1997 to 2016. Our empirical strategy uses the first difference GMM and system GMM model. To ensure the effectiveness of GMM estimation, we also conduct the FE and POLS estimation as a comparison. Following [Halkos and Paizanos \(2013\)](#)'s method, we further estimate the long-run effects of each independent variable on the basis of estimated short-run coefficients. Next, we examine whether our baseline findings remain robust to a number of alternative specifications and sample divisions.

Our estimates indicate that there is a significant long-run U-shaped relationship between urbanization and urban-rural electricity consumption inequality. The estimated turning point arrives at the urbanization level of around 60%. Our estimates also show that both the increasing urban-rural income gap and the urban-rural education gap will exacerbate urban-rural electricity consumption inequality. However, the inequality is negatively associated with infrastructure construction funds and fuels' price.

To eliminate the bias caused by the unobserved regional heterogeneity, we next explore how the U-shaped relationship changes in response to the regional heterogeneity. The heterogeneity analysis investigates the impacts of geographic and economic differences between China's provinces on the baseline

U-shaped relationship. There are two interesting findings: firstly, the colder northern region's turning point arrives later than the south. Secondly, the baseline U-shaped relationship between urbanization and urban-rural electricity consumption inequality is valid for developed eastern regions, while for the undeveloped midwest, the relationship is not nonlinear but linearly negative. As an extension, we lastly study the mechanism underlying the baseline U-shaped relationship. To clarify the mechanism, we carry out an analysis introducing the scale effect and efficiency effect. We find that the scale and efficiency effect change over urban and rural areas, and their interaction is probably a mechanism for the U-shaped relationship between urbanization and urban-rural electricity consumption inequality.

This study contributes to the literature on energy consumption inequality. Existing literatures have long focused on the measurement of energy disparity and tell us very little about the underlying causes of the disparity. Also, this study contributes to the literature on China's urban-rural inequality. Most of relevant studies analyze urban-rural inequality in terms of income and monetary measures. Our approach here provides another analytical methodology for urban-rural inequality from the perspective of electricity consumption. This paper is, to our knowledge, the first in the literature to focus specially on China's urban-rural electricity consumption inequality.

The rest of the paper proceeds as follows. Section 2 reviews the existing literature. Section 3 introduces the model and data. Section 4 carries out the empirical estimation and robustness check. Section 5 performs a heterogeneity analysis analyzing how our results change in response to regional geographical and economic differences. Section 6 explores the mechanism underlying our baseline U-shaped relationship. Section 7 concludes.

2 Literature review

This paper is related to several literatures. Most closely related is work studying the determinants of residential energy consumption. The existing literature points to mainly four factors (economy, demography, living condition and environmental attitudes) that impact on energy consumption. Firstly, economic factors are the most significant determinants, and among them, GDP per capita, income level and energy price have empirically been examined to be closely associated with residential energy consumption (Nesbakken, 1999; Galeotti et al, 2006; Druckman and Jackson, 2008). Secondly, other work has emphasized the importance of demographic factors in explaining energy users' consumption. Kinnear et al (1974), Yamasaki and Tominaga (1997), Metin Ozcan (2013) and Frederiks et al (2015), for example, have empirically examined the importance of household size, educational attainment and age structure. Thirdly, living condition factors affect residential energy consumption through different geographical features and house areas. For example, Brounen and Kok (2011) analyze the micro data of 3000 Dutch households' energy consumption, and find that household energy consumption increases

with house areas. Fourthly, environmental attitudes are also found to play a crucial role. Sapci and Considine (2014) find that attitudes about environmental issues are associated with lower energy consumption using a data set of electricity use by 612 households in Wyoming. Several studies have shown that environmental attitudes and environmental behavior are related to people's values and values can serve as guiding principles in people's energy behavior (Rokeach, 1973; Karp, 1996; Stern, 2000).

Another strand of literature, most specifically related to our measurement of urban-rural electricity consumption inequality, studies how to estimate energy consumption equity. Existing studies primarily use three measurement methods: Lorenz curves and Gini coefficients, Theil index and Atkinson index. Jacobson et al (2005) firstly extend the application of Lorenz curves and Gini coefficients to energy consumption equity. Then Wu et al (2017) apply this method to assess energy consumption inequality in rural China, and find that the overall inequality of energy consumption and expenditure varies greatly in terms of energy type, end-use demand, regions and climatic zones. Work by Pakrooh et al (2020) introduces the Theil index into the energy consumption equity analysis and concludes that the world's energy consumption inequality is decreasing with years. For the first time, Schlör et al (2013) use the Atkinson index as an analytical tool to calculate the distribution of energy consumption and use the epsilon parameter³ to explicitly reveal the inequality aversion of society. Later studies show that applying the Atkinson index could make a significant contribution to science and policy debates on energy equity.

Our findings also relate to the literature investigating the relationship between urbanization and energy consumption. Previous research has shown conflicting results, suggesting that the relationship between urbanization and energy use is complex. On the one hand, the scale effect of urbanization increases residents' energy consumption through the development of industrialization. The industrialization in the process of urbanization shifts residents' energy behavior from a low-energy mode to a high-energy mode (Jones, 1991; Parikh and Shukla, 1995; Cole and Neumayer, 2004). For example, York (2007) concludes that urbanization contributes substantially to changes in energy consumption by analyzing data for fourteen foundational European Union Nations covering the period 1960–2000. On the other hand, Poumanyong and Kaneko (2010) investigate the effects of urbanization on energy use and CO₂ emissions using the STIRPAT model and a balanced panel dataset of 99 countries over the period 1975–2005, and find that urbanization decreases energy use in some development stages. Many other studies share the similar finding and attribute such negative relationship to the improving energy efficiency along with urbanization (Pucher, 1990; Liddle, 2003; Chen et al, 2008).

In summary, previous studies just focus on estimating energy inequality and investigating urbanization's effect on energy consumption. There is no empirical literature combining the two subjects, which specially studies the

³Epsilon defines how sensitively the Atkinson index should react to income inequalities.

relationship between urbanization and electricity consumption inequality, not to speak of the urban-rural electricity consumption inequality.

3 Framework and data

3.1 Framework

In this paper, we explore the relationship between urbanization and urban-rural electricity consumption inequality using a modified STRIPAT model which was first proposed by Rosa and Dietz (1998). Because the urban-rural electricity consumption inequality, as our research subject, does burden China's urban and rural areas with different environmental pressures, it is reasonable to use the STRIPAT model to guide our analysis, which relates environmental pressures (or environmental impacts) to population, affluence (GDP per capita) and technology. The environmental pressure loaded by urban and rural China is

$$E_{irt} = a_1 P_{irt}^{b_1} A_{irt}^{c_1} e_{irt} \quad (1)$$

$$E_{iut} = a_2 P_{iut}^{b_2} A_{iut}^{c_2} e_{iut} \quad (2)$$

where E_{irt} and E_{iut} is the electricity consumption of province i at times t for rural and urban areas; P and A denote population and affluence, respectively; e represents the random error term and includes the technology shock, which accords with previous work⁴; a is the constant terms; and b and c are the elasticities of P and A . To capture urban-rural electricity consumption inequality, we take the ratio of urban area's electricity consumption per capita to the rural

$$\frac{E_{iut}/P_{iut}}{E_{irt}/P_{irt}} = a \left(\frac{P_{iut}}{P_{irt}} \right)^{b_1-1} \left(\frac{A_{iut}}{A_{irt}} \right)^{c_1} e_{it} * (P_{iut}^{b_2-b_1})(A_{iut}^{c_2-c_1}) \quad (3)$$

where $\frac{E_{iut}/P_{iut}}{E_{irt}/P_{irt}}$ represents the urban-rural electricity consumption inequality; $\frac{P_{iut}}{P_{irt}}$ is the ratio of urban population to the rural, which can also be seen as an indicator of urbanization level; $\frac{A_{iut}}{A_{irt}}$ is the ratio of urban affluence to the rural, standing for the urban-rural income inequality; and the last two terms, $(P_{iut}^{b_2-b_1})(A_{iut}^{c_2-c_1})$, can be regarded as other socioeconomic shocks on electricity consumption, which explain why elasticities of P and A for the urban and rural are different. For example, China's rural areas are often with more elderly dwellers because the young move to cities for jobs, and the elderly have a stronger demand of electricity consumption for heating and lighting. Thus, this demographic difference between urban and rural areas may produce different elasticities. For convenience of analysis, we incorporate such socioeconomic

⁴See Rosa and Dietz (1998): the technology term actually incorporates not only technology as it is usually conceived but also social organization, institutions, culture, and all other factors affecting human impact on the environment other than population and affluence, so it can be modeled as a residual term.

shocks into X_{it} , so Eq.(3) can be briefly expressed as

$$Ineq_{it} = aUrba_{it}^b Inco_{it}^c X_{it}^d e_{it} \quad (4)$$

where $Ineq_{it}$ is the electricity consumption inequality of province i at times t ; $Urba$ is the urbanization level; $Inco$ is the urban-rural income inequality; and X denotes other socioeconomic factors affecting electricity inequality. In addition, because geographical, economic and demographic features all vary across China's provinces, we introduce v_i to capture province-specific characteristics. Following previous work⁵, all variables are in logarithmic form to eliminate possible heteroscedasticity.

$$\ln Ineq_{it} = b \ln Urba_{it} + c \ln Inco_{it} + d \ln X_{it} + v_i + \ln e_{it} \quad (5)$$

Eq.(5) can be transformed into a dynamic model by adding lags of the dependent variable to the right side of Eq.(5). Compared with static models, the dynamic model is chosen because its advantage in estimating both long-run and short-run elasticities.

$$\ln Ineq_{it} = a \ln Ineq_{i(t-1)} + b \ln Urba_{it} + c \ln Inco_{it} + d \ln X_{it} + v_i + \ln e_{it} \quad (6)$$

Many empirical studies⁶ have found an Environmental Kuznets Curve (EKC) relationship between urbanization and environmental impacts. Intuitively, we further modify the model by adding the quadratic term of urbanization to capture the potential non-linear relationship between urbanization and electricity consumption inequality. Finally, the below Eq.(7) will guide our following empirical analysis.

$$\ln Ineq_{it} = a \ln Ineq_{i(t-1)} + b_1 \ln Urba_{it} + b_2 (\ln Urba_{it})^2 + c \ln Inco_{it} + d \ln X_{it} + v_i + \ln e_{it} \quad (7)$$

3.2 Data

The dataset is a balanced panel which contains observations of 30 China's provinces⁷ over periods of 1997-2016, with a sample of 600 observations. The provincial level data are obtained from China Statistical Yearbook, China City Statistical Yearbook, China Regional Statistical Yearbook, China Industrial Statistical Yearbook, China Energy Statistical Yearbook and Statistical Yearbooks of each province. Especially, the data of provinces' yearly electricity consumption are collected from each provinces' annual balanced sheet of electricity consumption, which annually records urban and rural electricity consumption of all China's regions and provinces. In addition, all economic variables are deflated based on the CPI of 1997.

⁵For corresponding literatures, see (Narayan and Smyth, 2008; Ozturk et al, 2010; Liddle, 2015).

⁶For example, Martínez-Zarzoso and Maruotti (2011) find an inverted-U shaped relationship between urbanization and CO2 emissions in developing countries from 1975 to 2003.

⁷Taiwan, Hong Kong, Macau and Tibet are not included in our analysis because of data unavailability.

Table 1 Descriptive statistics of variables

Variable	Definition of the variables(unit)	Obs	Mean	Std.Dev	Min	Max
Electricity (theil)	Urban-rural electricity consumption gap measured by theil index	600	.111	.142	0	.754
Electricity (ratio)	Urban-rural electricity consumption gap measured by ratio	600	.79	.6	.002	2.676
Urbanization	The ratio of urban population to rural's	600	1.216	1.357	.163	8.622
Income (ratio)	Urban-rural income gap measured by ratio	600	2.904	.611	1.623	5.498
Education	The ratio of urban average schooling years to rural's	600	1.347	.134	1.014	2.649
Infrastructure	The funds on infrastructure construction	600	281.738	300.059	1.237	2125
Fuel	The change of fuel price	600	.052	.084	-.165	.569
Climate1	Extremely cold areas	600	.167	.373	0	1
Climate2	Cold areas	600	.367	.482	0	1
Climate3	Hot summer and warm winter areas	600	.267	.443	0	1
Climate4	Hot summer and warm winter areas	600	.133	.34	0	1
Climate5	Temperate areas	600	.067	.25	0	1

To measure the dependent variable, China's provincial urban-rural electricity consumption inequality, we introduce the Theil index. The Theil index is a popular measurement for inequality, which was firstly proposed by [Sastry and Theil \(1969\)](#). We also take the ratio of urban electricity consumption per capita to the rural as another measurement for urban-rural electricity consumption inequality, which learns from [Dong and Hao \(2018\)](#)'s measurement of China's provincial income disparity. The two measurements have been broadly used in the research on measuring the economy and society inequality. We will use the Theil index in the baseline estimation and then use the ratio form for the robustness checks. The calculation methods of the Theil index and ratios are

$$Theil_{it} = \sum_{j=1}^n \left(\frac{Y_{jt}}{Y_{it}} \right) \left(\ln \frac{Y_{jt}}{Y_{it}} - \ln \frac{X_{jt}}{X_{it}} \right) \quad (8)$$

$$Ratio_{it} = \frac{e_{iut}}{e_{irt}} \quad (9)$$

where Y_{it} is the total electricity consumption of province i at times t ; X_{it} is the total population of province i at times t ; Y_{jt} is the electricity consumption of area j at times t ; X_{jt} is the population of area j at times t ; and it represents urban areas if $j=1$ and rural areas if $j=2$; e_{iut} and e_{irt} is the urban and rural electricity consumption per capita at times t in province i . In addition, for ratios, some of its logarithmic term is below 0 because some provinces' rural electricity consumption per capita will exceed the urban at a certain time. To better depict the inequality after the rural's outstripping the urban, we take the absolute value of these ratios.

The key independent variable, the urbanization level, is proxied by four indicators: the ratio of urban population to the rural, region's urban population to total population, the proportion of region's non-agricultural population and road network density. The first one is set by our modified STRIPAT model and will be used in the baseline estimation. The latter three variables are the alternative specifications of urbanization, which will be employed in the robustness checks. In addition, we also measure urban-rural income inequality by the ratio of urban GDP per capita to the rural .

Besides the above variables directly derived from the STRIPAT model, this paper includes other socioeconomic variables affecting urban-rural electricity consumption inequality. We firstly consider urban-rural education gaps because energy users' energy consumption patterns relatively depend on their human capital level. The urban-rural education gap is defined as the ratio of urban residents' average years of school attainment to the rural⁸; Secondly, the annual provincial infrastructure construction funds, which include investments on electricity facilities, can improve the electricity production and power transmission efficiency. And this improvement will further relieve region's electricity disparity, especially for the rural China. Thus we incorporate infrastructure construction funds into the analysis; Thirdly, as the substitute of electricity consumption, the price of fuels is closely related to electricity price, and therefore affects urban and rural electricity consumption. So the change of fuels' price is taken as another independent variable; Lastly, we are also interested in how climates across China's regions influence local urban-rural electricity consumption inequality. To investigate this relationship, this paper divides China into five zones according to their specific climatic features, and each climatic zone will be treated as a binary variable in the estimation⁹. Table 1 reports the descriptive statistics of all these variables.

To intuitively display the relationship between urban-rural electricity consumption inequality and urbanization, we picture the scatter diagram of electricity consumption inequality against the urbanization rate and draw a fitness line in Figure 2. As it shows, the line of best fit has a U-shaped curve and it implies that urban-rural electricity consumption inequality has a U-shaped relationship with urbanization. To more precisely investigate the relationship between urbanization and urban-rural electricity consumption inequality, some intricately detailed estimation will be employed in the following section.

4 Empirical Results

4.1 Baseline results

The above Eq.(7) guides our following empirical analysis and it is a dynamic panel model. To relieve potential endogenous problems, this paper also employs the first difference GMM (Arellano and Bond, 1991) and system GMM model (Arellano and Bover, 1995; Blundell and Bond, 1998), which are for "small T and large N" panel models. The first difference GMM can eliminate fixed effects and avoid endogenous problems by taking variables' lag-terms as the instrumental variables. However, problems of weak instruments will arise as a result of too many instrumental variables if the time series T is large enough. Based

⁸Following the popular method of calculating each province's education years from China Regional Statistical Yearbook, the corresponding relationship between school years and school attainment is: no schooling equals 0 years, primary schooling equals 6 years, junior high schooling equals 9 years, senior high schooling equals 12 years, college equals 16 years, and postgraduate and above equals 16 years.

⁹Based on the work of Wu et al (2017), China's five climatic zones are extremely cold areas, cold areas, hot summer and cold winter areas, hot summer and warm winter areas, and temperate areas. For more details, see Appendix.

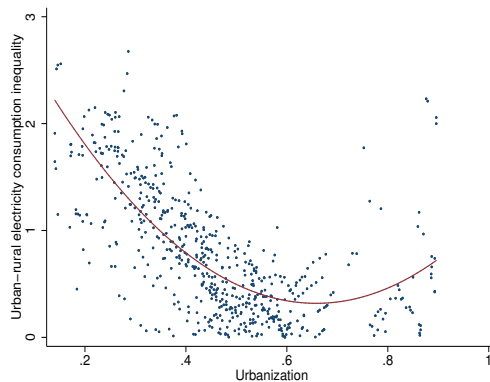


Fig. 2 The scatter diagram and fitting curve

on a new assumption that the first differences of instrumental variables are not correlated with the fixed effects, [Arellano and Bover \(1995\)](#) build system GMM model which simultaneously estimates the original equation and first differencing equation. Such treatment can reduce some bias of using the first difference GMM model. The system GMM estimator also allows the introduction of more instruments, which can dramatically improve estimation efficiency ([Roodman, 2009](#)).

Before the empirical estimation, a pre-test of unit root test is conducted to make sure that the series of panel data are stationary. We conduct three types of unit root tests: [Perron \(1988\)](#), [Levin et al \(2002\)](#) and [Im et al \(2003\)](#), to examine whether variables are stationary at levels. The three tests all have the null hypothesis that all panels contain a unit root. Table 2 shows that all variables are stationary at levels whether with the intercept or with the intercept and trend using the LLC, IPS and P.person tests. Then we can safely investigate the long relationship between the dependent variable and independent variables.

Table 3 reports the result of benchmark estimation which uses the Theil index as the dependent variable. As noted previously, the first difference GMM and system GMM methods are employed to relieve potential endogenous problems and corresponding estimation results are shown in column (5)-(8). To ensure the effectiveness of GMM estimation, we also conduct the FE and POLS estimation as a comparison, which are reported in column (1)-(4).

The validity of GMM estimates is determined by the Sargan test and AR (2) test. The results of the Sargan tests and Arellano-Bond tests are reported on the bottom of Table 3. The Sargan test is performed to examine the effectiveness of the instrument variables used in GMM models and the null hypothesis of this test assumes that the instrumental variables are valid. The Arellano-Bond test is conducted to verify the existence of second-order autocorrelation in residuals, which is a presumption of the GMM model. The results of Sargan tests reveal that the instruments set is valid and AR(2) test results accept the

Table 2 The unit root test

	Intercept			Intercept and trend		
	LLC	IPS	Pperson	LLC	IPS	Pperson
Electricity	-3.176***	-1.436*	81.423**	-3.679**	-3.911**	99.091**
L.Electricity	-3.414***	-1.605*	78.425*	-3.313**	-4.280**	99.983**
Urbanization	-5.494***	-5.332***	179.958***	-11.773**	-7.136**	174.058**
<i>Urbanization</i> ²	-23.890***	-18.671***	230.953***	-17.783**	-7.324**	139.944**
Income	-4.105***	-2.710***	154.727***	-4.652**	-4.517**	165.233**
Education	-10.129***	-9.847***	211.909***	-11.093**	-9.868**	217.162**
Infrastructure	-3.384***	-2.996***	114.669***	-5.380**	-4.136**	121.984**
Fuel	-18.473***	-17.010***	594.245***	-15.136**	-13.901**	504.351**

Note: L.Electricity represents the lag of the dependent variable; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

null hypothesis of no autocorrelation of the second order in residuals, which verifies the validity of GMM estimation. Moreover, Bond (2008) concludes that a consistent and reasonable GMM estimator should not be significantly higher than the OLS estimator or significantly lower than the FE estimator. Our results show that most of the GMM estimators are slightly lower than the POLS's and higher than the FE's, which further confirms the effectiveness of our GMM estimation.

As baseline estimation result shows in Table 3, there is a significant U-shaped relationship between urbanization and urban-rural electricity consumption inequality, and the estimated turning point arrives at the urbanization level of 61.34%, 62.36%, 59.81%, and 59.57% for column (5), (6), (7) and (8). This finding indicates that the urban-rural electricity consumption inequality shifts from a downward trend to an upward trend when the urbanization rate is around 60%. Our estimates also show that the increase of urban-rural income gap and urban-rural education gap will exacerbate urban-rural electricity consumption inequality. However, the inequality is negatively associated with infrastructure construction funds and the rise of fuel price. Moreover, column (2), (4), (6) and (8) all adds the interaction term of the fuel price's change and urban-rural income gap, and the results suggest that the rise of fuel price will enlarge the positive effect of urban-rural income gap on urban-rural electricity consumption inequality.

It should be noted that the above empirical result is based on the autoregressive distributed lag model (ADL)¹⁰ and its estimated coefficients reflect the short-run effects. Following Halkos and Paizanos (2013), we further estimate the long-run effects of each independent variable by making the short-run coefficients be divided by the difference between 1 and the coefficient of lagged

¹⁰In the standard ADL model, the regressors may include lagged values of the dependent variable and current and lagged values of one or more explanatory variables. For our empirical model in this paper, the number of the lagged terms of independent variables is automatically determined by our setting of GMM.

Table 3 The baseline estimation results

Dep. Var: Theil index	POLS		FE		Diff-GMM		Sys-GMM	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
L.Electricity	0.803*** (0.037)	0.797*** (0.037)	0.726*** (0.028)	0.716*** (0.028)	0.732*** (0.048)	0.719*** (0.045)	0.796*** (0.038)	0.795*** (0.036)
Urbanization	-0.060** (0.026)	-0.059** (0.026)	-0.104*** (0.036)	-0.108*** (0.036)	-0.096* (0.050)	-0.103** (0.052)	-0.066** (0.027)	-0.062** (0.026)
<i>Urbanization</i> ²	0.079*** (0.017)	0.078*** (0.017)	0.106*** (0.018)	0.103*** (0.018)	0.104*** (0.025)	0.102*** (0.024)	0.083*** (0.017)	0.080*** (0.016)
Income	0.159*** (0.058)	0.103 (0.065)	0.296*** (0.085)	0.229*** (0.088)	0.294*** (0.100)	0.227** (0.102)	0.160*** (0.056)	0.104* (0.063)
Education	0.197*** (0.056)	0.186*** (0.052)	0.180* (0.099)	0.160 (0.099)	0.175*** (0.067)	0.157** (0.064)	0.215*** (0.064)	0.180*** (0.050)
Infrastructure	-0.014** (0.006)	-0.016** (0.006)	-0.026* (0.013)	-0.030** (0.013)	-0.028** (0.012)	-0.031*** (0.011)	-0.013** (0.006)	-0.016*** (0.006)
Fuel	-0.449*** (0.104)	-1.620*** (0.576)	-0.498*** (0.103)	-1.894*** (0.466)	-0.499*** (0.128)	-1.921*** (0.603)	-0.443*** (0.101)	-1.621*** (0.558)
<i>Fuel * income</i>		0.409** (0.190)		0.485*** (0.158)		0.495*** (0.190)		0.409** (0.184)
Climate1	0.078** (0.030)	0.082** (0.030)	No	No	No	No	0.083*** (0.030)	0.084*** (0.030)
Climate2	0.033 (0.028)	0.035 (0.028)	No	No	No	No	0.036 (0.028)	0.036 (0.028)
Climate3	0.050* (0.028)	0.055* (0.028)	No	No	No	No	0.054* (0.028)	0.056** (0.027)
Climate4	0.014 (0.024)	0.014 (0.024)	No	No	No	No	0.018 (0.023)	0.015 (0.024)
Constants	-0.317*** (0.079)	-0.234*** (0.079)	-0.289 (0.201)	-0.167 (0.203)			-0.348*** (0.094)	-0.228*** (0.077)
Sargan					0.742	0.347	0.382	0.371
A-B test for AR (1)					0.000	0.000	0.000	0.000
A-B test for AR (2)					0.937	0.863	0.966	0.913
<i>N</i>	570	570	570	570	540	540	570	570
<i>R</i> ²	0.911	0.912	0.858	0.860				

Note: Standard errors in parentheses; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

dependent variable. Because the estimation using GMM models is more accurate and reliable, we just estimate the long-run estimates using the short-run coefficients of first difference GMM and system GMM. The results of long-run effects are reported in Table 4 and the following discussion is based on the long-run estimates.

The long-run coefficients on urbanization underlying column (1), (2), (3) and (4) in Table 4 indicate that there is a significant long-run U-shaped relationship between urbanization and urban-rural electricity consumption inequality. The estimated turning point arrives at the urbanization level of around 60% (61.33%, 62.38%, 59.82% and 59.56% for column (1), (2), (3) and (4), respectively), which is quite similar to the estimated short-run coefficients in Table 3. For this U-shaped relationship, we find evidence of two opposite impacts from the scale effect and efficiency effect emerging in urbanization. The scale effect is the pull effect of urbanization to both urban and rural residents' electricity consumption, and the efficiency effect is the differentiated shocks to urban and rural electricity demand in the late course of urbanization, which

is caused by the improved power efficiency along the urbanization. Specifically, in the early course of urbanization, the scale effect is dominant and it has a stronger push to rural electricity consumption than to the urban, which helps to decrease the gap between urban and rural electricity consumption. In line with such idea, the data from the China National Energy Administration shows that rural electricity consumption per capita, especially after 2000, grows much more rapidly than urban¹¹. In the late course of urbanization, urban environment and resources are burdened with much pressure caused by the large population inflow. To relieve city's pressure, the predominant efficiency effect drives urban electricity consumption more efficiently while rural electricity consumption keeps surging, thus eventually leaving some rural areas' electricity consumption overtaking urban's. We will further detailedly investigate the scale and efficiency effect to urban and rural electricity consumption in section 6.

In the following discussion, our preferred specification is that of column (4), which is based on the system GMM and absorbs both the interaction term as well as the climate effects. Firstly, the estimated long-run coefficient on the urban-rural income gap is significantly positive, showing that a 1% decline in the ratio of urban income per capita to the rural decreases the urban-rural electricity consumption inequality by 0.507%. This finding is in accordance with the "Energy ladder" theory (Reddy and Reddy, 1994) which holds that energy users shift from one inferior fuel source (firewood) to the superior source (electricity) with an increase in their economic status. Secondly, the urban-rural electricity consumption inequality will decrease by 0.878% if the gap between urban and rural education decreases by 1%. For this conclusion, we think rural energy users will shift their energy consumption pattern from a high-energy-consumption way to an energy-efficient way when they are better educated, and thus these better educated rural energy users gradually replace their traditional biomass consumption with relatively cleaner electricity consumption, which helps to decrease the urban-rural electricity consumption inequality.

Thirdly, the infrastructure construction fund is negatively associated with the urban-rural electricity consumption inequality and a 1% increase in the infrastructure construction fund gives rise to a 0.078% decline in the inequality. Fourthly, because the fuel price's change is not in logarithmic form, the coefficient on it can not be interpreted as the elasticity. The estimated coefficient on the fuel price's change suggests that the rise of fuels' price could decrease the urban-rural electricity consumption inequality. For this negative relationship, Nesbakken (1999) finds fuels' price is the dominant factor affecting households' energy consumption, so when fuels' price rises, China's rural households who are used to consuming coals will switch to electricity consumption, an alternative energy of fuels. Moreover, the significantly positive coefficient on the interaction term indicates that the urban-rural income gap

¹¹ From 2008 to 2011, China's rural residential electricity consumption experiences an average 12% to 14% growth yearly, which is higher than that of the urban (8.9%-11.3%). The ratio of urban electricity consumption per capita to the rural has declined from 1.6 (2008) to 1.4 (2010), and then to 1.2 (2012).

may enlarge the effect of fuel's price on urban-rural electricity consumption inequality. About the underlying causality, we guess that the larger urban-rural income gap implies the rural energy users are relatively poorer and therefore they are more sensitive to the rising fuels' price.

Lastly, the climate impact on urban-rural electricity consumption inequality, which is eliminated by the first difference GMM, is estimated by the system GMM, and the results show that only the coefficients on climate1 and climate3 are significant. Climate1 represents regions with extremely cold weather and the coefficient on it is positive at 1 percent level. Regions with extremely cold weather are mainly located in the northernmost and westernmost of China and the rural areas of these regions are either undeveloped or remote. For these regions' rural areas, the energy system lags far behind the urban. For example, 39.8 thousand rural residents of Qinghai, one of the extremely cold provinces, are still inaccessible to electricity consumption in 2015 and their main life energy comes from biomass energy. Due to these rural areas' lagged energy system, the urban-rural electricity consumption inequality of these extremely cold areas is greater. In addition, it's surprising that the coefficient on climate3, the climate of hot summer and cold winter, is also significantly positive and the underlying reason is waiting for future investigation.

Table 4 The long-run estimates of explanatory variables by GMM

Dep. Var: Theil index	Diff-GMM		Sys-GMM	
	(1)	(2)	(3)	(4)
Urbanization	-0.358 *	-0.367 **	-0.324 **	-0.302 **
<i>Urbanization</i> ²	0.388 ***	0.363 ***	0.407***	0.390 ***
Income	1.097 ***	0.808 **	0.784***	0.507 *
Education	0.653 ***	0.559 **	1.054***	0.878 ***
Infrastructure	-0.104**	-0.110***	-0.064 **	-0.078 ***
Fuel	-1.862 ***	-6.836***	-2.172 ***	-7.907 ***
<i>Fuel * income</i>		1.762 ***		1.995 **
Climate 1			0.407***	0.410***
Climate 2			0.176	0.176
Climate 3			0.265 *	0.273 **
Climate 4			0.088	0.073

Note: Standard errors in parentheses; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

4.2 Robustness checks

In the following analysis, we will examine whether our baseline findings remain robust to a number of alternative specifications and sample divisions.

4.2.1 Another measurement of electricity inequality

So far, we have measured the urban-rural electricity consumption inequality by the Theil index. Another measurement, the ratio of urban electricity consumption per capita to the rural, is employed to assess whether baseline results remain. The set of the ratio is

$$Ratio_{it} = \frac{e_{iut}}{e_{irt}} \quad (10)$$

where e_{iut} and e_{irt} is the urban and rural electricity consumption per capita at times t in province i . In addition, for ratios, some of its logarithmic term is below 0 because some provinces' rural electricity consumption per capita will exceed the urban at a certain time. To better depict the inequality after the rural's outstripping the urban, we take the absolute value of these ratios.

Table 5 reports the results of using the ratio of urban electricity consumption per capita to the rural as the dependent variable. The results are generally similar to our baseline results, and the U-shaped relationship between urbanization and urban-rural electricity consumption inequality remains unchanged. Specifically, the estimated turning point (column 3 and 4) is 64.69% and 65.84%, which quite approximates the baseline estimation. In addition, the estimated coefficients suggest that urban-rural electricity consumption inequality is positively associated with the urban-rural income inequality and education gap, and negatively associated with infrastructure construction funds and the change of fuel's price, which accords with our baseline findings.

4.2.2 Other measurements of urbanization

Recall that our model defines urbanization as the ratio of urban population to the rural. Urbanization can also be measured by the ratio of region's urban population to region's total population and the proportion of region's non-agricultural population, which are two popular measurements of urbanization in relevant studies. Additionally, we use province's road network density as another alternative of urbanization. The validity of road density has been verified by many researches on urbanization. For example, [Rudel and Richards \(1990\)](#) find that regions with better traffic are often with more rapid urbanization while less developed transportation system impedes regions' urbanization. Next, we consider whether our baseline results are sensitive to these different measurements of urbanization.

The estimates in Table 6 are divided into three: column 1 and 2, 3 and 4, and 5 and 6 replace urbanization with the ratio of region's urban population to region's total population, the proportion of region's non-agricultural

Table 5 The robustness check 1

Dep. Var:Urban-rural ratio	POLS	FE	Diff-GMM	Sys-GMM
	(1)	(2)	(3)	(4)
L.Theil	0.758*** (0.037)	0.710*** (0.025)	0.734*** (0.045)	0.756*** (0.036)
Urbanization	-0.021*** (0.006)	-0.031*** (0.008)	-0.023* (0.013)	-0.021*** (0.006)
<i>Urbanization</i> ²	0.015*** (0.004)	0.021*** (0.004)	0.019** (0.007)	0.016*** (0.005)
Income	0.022 (0.015)	0.052*** (0.019)	0.041 (0.026)	0.023 (0.015)
Education	0.059*** (0.017)	0.063*** (0.022)	0.053*** (0.020)	0.059*** (0.016)
Infrastructure	-0.004** (0.002)	-0.005 (0.003)	-0.008** (0.004)	-0.004** (0.002)
Fuel	-0.117*** (0.027)	-0.127*** (0.023)	-0.129*** (0.029)	-0.116*** (0.026)
Climate zones	Yes	No	No	Yes
Constants	-0.074*** (0.025)	-0.101** (0.044)		-0.076*** (0.024)
Sargan			0.337	0.931
A-B test for AR (1)			0.000	0.000
A-B test for AR (2)			0.660	0.665
<i>N</i>	570	570	540	570
<i>R</i> ²	0.921	0.881		

Note: L.Theil represents the lag of the dependent variable; ***, **; Standard errors in parentheses; * represent significance at the 1%, 5% and 10% levels, respectively.

population and road network density, respectively. Each of the three is based on the first difference GMM and system GMM. The coefficients of column 1 to 6 show that the U-shaped relationship does not change when we employ different measurements of urbanization.

4.2.3 Sample divisions

Our baseline results imply that the turning point of the U-shaped relationship arrives at the urbanization rate of 60%. To assess how urban-rural electricity consumption inequality changes before and after the turning point, we split sample periods into two parts: 1997-2005 and 2006-2016. The reason why we select the year 2005 as the breakpoint is that some eastern provinces' urbanization rate has approximately reached 60%¹², which is around the turning

¹²In 2005, the urbanization rate of Jiangsu, Zhejiang, Guangdong is 56%, 57% and 61%, respectively.

Table 6 The robustness check 2

Dep. Var: Theil index	Urbanization rate		Non-agricultural population		Road density	
	Diff-GMM	Sys-GMM	Diff-GMM	Sys-GMM	Diff-GMM	Sys-GMM
	(1)	(2)	(3)	(4)	(5)	(6)
L.Electricity	0.733*** (0.050)	0.830*** (0.031)	0.795*** (0.033)	0.871*** (0.024)	0.791*** (0.028)	0.869*** (0.027)
Urbanization2	-2.569*** (0.767)	-1.577*** (0.392)				
Urbanization2 ²	2.523*** (0.724)	1.434*** (0.363)				
Nonagricultural			-2.428*** (0.788)	-0.896** (0.406)		
Nonagricultural ²			0.352*** (0.114)	0.129** (0.058)		
Road					0.063 (0.101)	0.057*** (0.019)
Road ²					0.036*** (0.013)	0.009*** (0.002)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Climate zones	Yes	Yes	Yes	Yes	Yes	Yes
Constants		0.175* (0.106)		1.303* (0.696)		0.126 (0.105)
Sargan	0.399	0.775	0.910	0.890	0.895	0.982
A-B test for AR (1)	0.000	0.000	0.000	0.000	0.000	0.000
A-B test for AR (2)	0.984	0.920	0.981	0.990	0.920	0.937
N	540	570	540	570	540	570

Note: L.Electricity represents the lag of the dependent variable; Urbanization2 is the ratio of urban population to total population; Nonagricultural is the ratio of nonagricultural population to total population; Road is the road density; Standard errors in parentheses; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

point. In addition, this treatment can also ensure that two new samples all have enough observations for our GMM estimation.

Table 7 reports the results of sample divisions. The coefficients of sample 1997-2005 show that a 1% increase in urbanization would decrease urban-rural electricity consumption inequality by 0.547%-0.608%. In contrast, for sample 2006-2016, urban-rural electricity consumption inequality only declines by 0.075%-0.093%, which is about one-sixth of sample 1997-2005. We suppose that in sample 2006-2016, some eastern provinces' rural electricity consumption per capita outweighs the urban, leading its coefficients much smaller than that of sample 1997-2005. In other words, urbanization's overall decreasing effect on urban-rural electricity consumption inequality declines a lot after the relationship between urbanization and urban-rural electricity consumption inequality turning to be positive for some developed eastern provinces. Thus, the results of sample divisions indirectly confirm the U-shaped relationship between urbanization and urban-rural electricity consumption inequality.

Table 7 The robustness check 3

Dep. Var: Theil index	1997-2005		2006-2016	
	Diff-GMM	Sys-GMM	Diff-GMM	Sys-GMM
	(1)	(2)	(3)	(4)
L.Electricity	-0.105*** (0.021)	-0.101*** (0.023)	-0.135*** (0.034)	-0.076*** (0.030)
Urbanization	-0.547*** (0.032)	-0.608*** (0.035)	-0.075** (0.036)	-0.093** (0.038)
Income	1.092*** (0.078)	0.661*** (0.100)	0.495*** (0.185)	0.667*** (0.147)
Education	0.533*** (0.086)	0.699*** (0.083)	1.617*** (0.178)	1.361*** (0.215)
Infrastructure	-0.152*** (0.009)	-0.160*** (0.013)	0.012 (0.033)	-0.034 (0.026)
Fuel	0.332 (0.242)	-0.494 (0.329)	0.076 (0.425)	0.426* (0.221)
Climate zones	Yes	Yes	Yes	Yes
Constants		-0.125 (0.211)		-1.503*** (0.284)
Sargan	0.326	0.287	0.125	0.162
A-B test for AR (1)	0.005	0.004	0.002	0.001
A-B test for AR (2)	0.296	0.191	0.329	0.233
<i>N</i>	252	261	308	319

Note: Standard errors in parentheses; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

5 Heterogeneity analysis

One might be concerned that unobserved regional heterogeneity might bias our estimates. The regional heterogeneity includes geographical and economic differences between China's provinces. These differences are closely related to energy users' electricity consumption, and impact on urbanization's effect on the urban-rural electricity consumption inequality. We next explore in more detail how the U-shaped relationship responses to the regional heterogeneity.

5.1 Geographical differences

Energy users' consumption pattern closely relates to local geographic and climatic features. In the case of China, geographically, the separation between the north and south is defined by the border which runs along the Qingling Mountains from Sichuan through the southern Shaanxi Province eastwards along the Huai River ending in the Pacific. Generally, the south is warmer and

wetter in climate than the north. This explains why there is no heating system in southern China. Also, northern China is much richer in coal resources than the south. We suppose these differences in geographic and climatic features might affect locals' daily electricity consumption pattern. We then examine the impact of such geographical division on the relationship between urbanization and urban-rural electricity consumption inequality.

Given that grouped regressions based on the division will cause the loss of observations, this paper takes the varying coefficient model. This model helps to examine how regression coefficients change over regions with different geographical divisions without losing observations. The empirical specification is as follows

$$\ln Gap_{it} = (a_1 D_{1j} + a_2 D_{2j}) \ln Gap_{i(t-1)} + (b_1 D_{1j} + b_2 D_{2j}) \ln Urba_{it} + (c_1 D_{1j} + c_2 D_{2j}) (\ln Urba_{it})^2 \dots + \ln e_{it} \quad (11)$$

where,

$$D_{1j} = \begin{cases} 1 & South \\ 0 & Nonsouth \end{cases} \quad D_{2j} = \begin{cases} 1 & North \\ 0 & Nonnorth \end{cases}$$

The results are presented in Table 8 and the long-run estimates are reported in Table 9. The estimation of column 1 and 2, and 3 and 4 are for the south and north, respectively, and both are based on the first difference GMM and system GMM. The coefficients on urbanization and *urbanization*² imply that the U-shaped relationship remains significant for both the south and north. As the long-run estimates show, northern region's turning point arrives later than the south, specifically, it is around the urbanization rate of 71.48%-74.98% and the south is around 55.38%-59.61%.

The reason to interpret the different turning points of the south and north is: northern Chinese, especially in the rural, is more used to consuming coals for daily heating and cooking due to northern China's abundant coal supplies. For example, according to the National Energy Administration, more than 90% of northern rural areas still use bulk coals for winter heating in 2017. This means that northern energy users' transformation from the coal-consumption-habit to the electricity-consumption-habit might be much harder. Besides, backward power facilities of northern rural areas make the transformation more difficult. Southern China, by contrast, has weaker heating demand due to its warmer climate. Also, the relatively developed power facilities of southern rural areas make rural habitats more accessible to electricity consumption.

5.2 Economic differences

There are numerous empirical studies show that economic level is an important determinant of residential energy consumption patterns. So it is reasonable to suppose that regions with different economic levels may have differentiated relationship between urbanization and urban-rural electricity consumption inequality. To investigate such heterogeneity of regional economic disparity, we

divide China into two regions: the developed east and undeveloped midwest¹³. Based on that division, the developed east includes 12 provinces (areas), and the undeveloped midwest includes 18 provinces (areas). About the empirical specification, we follow the above varying coefficient models

$$D_{1j} = \begin{cases} 1 & \text{East} \\ 0 & \text{Noneast} \end{cases} \quad D_{2j} = \begin{cases} 1 & \text{Midwest} \\ 0 & \text{Nonmidwest} \end{cases}$$

The results are presented in Table 8. The estimation of column 5 and 6, and 7 and 8 are for the east and midwest, respectively, and both are based on the first difference GMM and system GMM. The Sargan test of the first difference GMM shows its P value is less than 0.1, indicating there is an over-identification of instrumental variables. So our following analysis relies on the results of the system GMM. Table 9 reports the long-run estimates.

As is clear from Table 9, the long-run coefficients in column 2 and 4 show that the U-shaped relationship between urbanization and urban-rural electricity consumption inequality is valid for developed eastern regions, while for the undeveloped midwest, the relationship is not nonlinear but negative linearly. This difference results from the economic disparity between eastern and midwestern regions. In particular, richer eastern regions' urbanization is much faster than midwestern regions¹⁴. For the undeveloped midwest, its urbanization has not yet come to the turning point and its urban-rural electricity consumption inequality decreases with the development of urbanization. However, for the developed east, the estimated long-run turning point is at the urbanization rate of 60.34%, which is very close to our baseline results (60%). This finding also points to the fact that eastern region's urbanization is ahead of other regions and some provinces' rural electricity consumption per capita has exceeded the urban.

6 Extensions

The most important finding so far is that there is a U-shaped relationship between urbanization and urban-rural electricity consumption inequality. However, it is empirically unclear how urbanization impacts on urban-rural electricity consumption inequality and what the mechanism underlying such U-shaped relationship. To address this question, we carry out an analysis introducing the scale effect and efficiency effect.

As we mentioned above, the U-shaped relationship can be theoretically explicated by two effects of urbanization: the scale and efficiency effect. Succinctly, the scale effect is the pull effect of urbanization to both urban and rural

¹³Eastern provinces are Hebei, Beijing, Tianjin, Shandong, Jiangsu, Shanghai, Zhejiang, Fujian, Guangdong, Hainan, Taiwan, Hong Kong and Macao. Others are midwestern provinces.

¹⁴According to China's National Bureau of Statistics, in 2018, the urbanization rate of some eastern provinces (Shanghai, Beijing, Tianjin, Guangdong, Jiangsu and Zhejiang) is more than 70%, while it is still below 50% for some western provinces (Yunnan, Gansu and Guizhou).

Table 8 Heterogeneity analysis

Dep. Var: Theil index	Economic differences		Geographical differences	
	Diff-GMM	Sys-GMM	Diff-GMM	Sys-GMM
	East		South	
	(1)	(2)	(5)	(6)
L.Electricity	0.694*** (0.081)	0.823*** (0.060)	0.665*** (0.109)	0.805*** (0.053)
Urbanization	-0.261*** (0.099)	-0.096** (0.044)	-0.269** (0.112)	-0.065 (0.041)
<i>Urbanization</i> ²	0.080*** (0.026)	0.103*** (0.025)	0.099*** (0.025)	0.118*** (0.028)
Income	0.550*** (0.204)	0.224 (0.145)	0.598*** (0.161)	0.299*** (0.106)
Education	0.262 (0.191)	0.151 (0.117)	0.110 (0.132)	0.102 (0.091)
Infrastructure	-0.028 (0.024)	-0.032*** (0.010)	-0.012 (0.021)	-0.027* (0.014)
Fuel	-0.603*** (0.194)	-0.480*** (0.156)	-0.541*** (0.176)	-0.453*** (0.126)
	Midwest		North	
	(3)	(4)	(7)	(8)
L.Electricity	0.570*** (0.082)	0.715*** (0.077)	0.628*** (0.064)	0.756*** (0.053)
Urbanization	-0.484*** (0.079)	-0.209*** (0.066)	-0.446*** (0.089)	-0.145*** (0.033)
<i>Urbanization</i> ²	0.036 (0.084)	0.041 (0.056)	0.057*** (0.019)	0.062*** (0.021)
Income	0.319 (0.262)	0.261* (0.139)	0.295 (0.183)	0.282** (0.113)
Education	0.108 (0.084)	0.192** (0.088)	0.211** (0.101)	0.202** (0.093)
Infrastructure	-0.018 (0.015)	-0.031** (0.014)	-0.005 (0.019)	-0.040*** (0.012)
Fuel	-0.399*** (0.149)	-0.389*** (0.149)	-0.463** (0.190)	-0.473*** (0.173)
Constants		-0.324* (0.171)		-0.348** (0.155)
Sargan	0.976	0.604	0.960	0.785
A-B test for AR (1)	0.005	0.003	0.006	0.003
A-B test for AR (2)	0.935	0.897	0.832	0.884
<i>N</i>	540	570	540	570

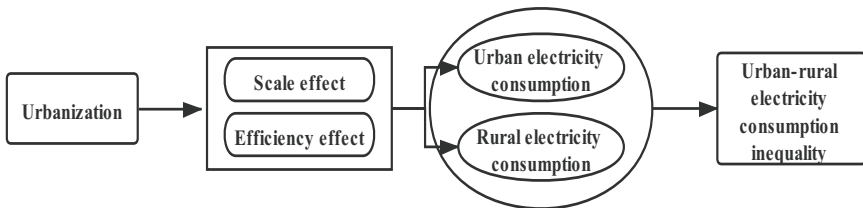
Note: Standard errors in parentheses; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

Table 9 The long-run estimates

Dep. Var: Theil index	Economic differences				Geographical Differences			
	Richer east		Undeveloped midwest		South		North	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Urbanization	-0.853 ***	-0.542**	-1.126 ***	-0.733 ***	-0.803**	-0.333	-1.199 ***	-0.594***
Urbanization ²	0.261***	0.582***	0.084	0.144	0.296 ***	0.605***	0.153 ***	0.254 ***
Income	1.797 ***	1.266	0.742	0.916 *	1.785 ***	1.533***	0.793	1.156 **
Education	0.856	0.853	0.251	0.674**	0.328	0.523	0.567 **	0.828**
Infrastructure	-0.092	-0.181***	-0.042	-0.109**	-0.036	-0.138*	-0.013	-0.164 ***
Fuel	-1.971***	-2.712 ***	-0.928 ***	-1.365***	-1.615 ***	-2.323***	-1.245 **	-1.939***

Note: Standard errors in parentheses; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

residents' electricity consumption, and the efficiency effect helps save electricity consumption by the improved energy efficiency. It should be noted that the two effects might change over the urban and rural due to China's urban-rural dual economic system¹⁵. About China's urban-rural dual economic system, two comments are in place. First, it limits the free move from the rural to urban by implementing a strict household registration system. Second, it widens the social and economic gap between rural and urban citizens by implementing different disposal mechanisms of resources¹⁶. According to Wei and Gong (2019), the "two systems" has undergirded not only China's industrialization, but also urbanization and rural-urban dynamics. So because of this system, urbanization may have differentiated effects on urban and rural energy consumption patterns. Then, we will analyze the scale and efficiency effect from the rural and urban perspective, independently. The relevant mechanism is shown in the below flow chart.

**Fig. 3** The mechanism flowchart

Firstly, we use electricity consumption per capita as the dependent variable and carry out regressions based on both the first difference GMM and system GMM for urban and rural samples, independently. Results are shown in Table

¹⁵Kuang and Liu (2012) note that China's rural-urban dual society system is instituted by its unique Hukou system. This system causes inequalities in social status between permanent urban and rural residents, and discrimination against rural-to-urban migrants is thus prevalent.

¹⁶Urban education and infrastructure are almost supported by national finance while a considerable part of rural funds is supported by themselves. This differentiated system further results in the urban-rural human capital gap and separates rural industry from urban industry.

10, and all estimations pass both Sargan and AR test. The most striking difference between the results of urban and rural samples is the magnitude of coefficients on urbanization. For urban samples, a 1% increase in urbanization rate raises electricity consumption per capita by 0.094%, reported in column 1. The corresponding coefficient for rural samples is 0.135%. This difference implies that there is a more rapid rise in rural electricity consumption along with urbanization. Some examples can be informative: according to the State Statistical Bureau, among 2008 to 2011, the yearly electricity consumption growth rate of rural areas is about 12%-14%, while it is about 8.9%-11% for urban areas. One way of seeing the difference in magnitude is that the scale effect of rural areas might be much bigger than urban areas and it predominates over the efficiency effect. To examine such conjecture, this paper will further quantify the scale and efficiency effect, and investigate their specific impacts on urban and rural electricity consumption.

Table 10 The electricity consumption of the urban and rural

Dep. Var: Electricity consumption	Urban		Rural	
	Diff-GMM	Sys-GMM	Diff-GMM	Sys-GMM
	(1)	(2)	(3)	(4)
L.Electricity	0.692*** (0.056)	0.822*** (0.028)	0.878*** (0.034)	0.978*** (0.021)
Urbanization	0.094** (0.042)	0.054*** (0.012)	0.135** (0.084)	-0.013 (0.033)
Other control variables	Yes	Yes	Yes	Yes
Constants		1.034*** (0.146)		0.111 (0.102)
Sargan	0.979	0.959	0.515	0.907
A-B test for AR (1)	0.010	0.016	0.000	0.000
A-B test for AR (2)	0.182	0.205	0.837	0.699
<i>N</i>	540	570	540	570

Note: Standard errors in parentheses; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

In order to estimate the scale and efficiency effect, we need to find ways to define the two effects at first. For the scale effect, Chang-hong et al (2006) conclude that the industrialization in the process of urbanization promotes electricity consumption. The reason is that citizens' electricity demand is boosted by the industrialization, especially the development of power equipment manufacturing industry. Meanwhile, power facilities are improved and thus energy users are more accessible to electricity consumption. So we define the scale effect as the industrialization rate, which is estimated as the ratio of industrial production to GDP. For the efficiency effect, it mainly refers to the technical advances and efficiency improvement. We define the efficiency effect

as the energy intensity, also known as the ratio of final energy consumption to GDP.

Following the empirical method of [Zhang \(2019\)](#), we introduce urbanization rate's interactions with the industrialization rate and energy intensity to depict shocks of scale and efficiency effect on urban and rural electricity consumption. The empirical specification is

$$Electricity_{it} = \alpha_0 + \alpha_1 Electricity_{i(t-1)} + \alpha_2 Urbanization_{it} + \alpha_3 Urbanization_{it} \times Industrial_{it} + Z + \epsilon_{it} \quad (12)$$

$$Electricity_{it} = \alpha_0 + \alpha_1 Electricity_{i(t-1)} + \alpha_2 Urbanization_{it} + \alpha_3 Urbanization_{it} \times Intensity_{it} + Z + \epsilon_{it} \quad (13)$$

where $Electricity_{it}$ is the electricity consumption per capita of region i at time t ; $Urbanization_{it}$ is the urbanization rate of region i at time t ; $Industrial_{it}$ and $Intensity_{it}$ are the industrial rate and energy intensity, respectively, and their interactions with urbanization are $Urbanization_{it} \times Industrial_{it}$ and $Urbanization_{it} \times Intensity_{it}$, which captures the scale and efficiency effect. It represents urban areas if $i = 1$ and rural areas if $i = 2$.

Table 11 reports the results of scale and efficiency effect. The estimates of column 1 to 4 show that the efficiency effect on urban and rural electricity consumption is opposite. In the case of urban electricity consumption, the coefficient on the interaction ($Urbanization_{it} \times Intensity_{it}$) is significantly negative, indicating that urbanization's pull effect on electricity consumption is vitiated by the improving energy efficiency. However, for rural electricity consumption, the corresponding coefficient is positive which suggests that the improving energy efficiency enlarges urbanization's positive impact on electricity consumption. The reason may be that to meet daily electricity demand like before, rural energy users now spend less as a result of the improving efficiency. Then the saved expenses are used for relatively advanced household entertainment electric appliances, like smart TV, stereo equipments and computers, and it eventually promotes rural household electricity consumption.

Column 5 to 8 show the results of scale effects for urban and rural areas. As for rural areas, the positive coefficient on the interactions ($Urbanization_{it} \times Industrial_{it}$) is significant at 5 percent, which is reported in column 7. This estimate points out the fact that the electricity consumption of China's rural energy users increases with the industrialization in the process of urbanization. In contrast, as column 6 shows, the interaction is negatively associated with urban electricity consumption. Maybe surprisingly, this negative coefficient suggests that the development of industrialization weakens urbanization's pull effect on electricity consumption. One explanation is that with the development of industrialization, urban habitants' lifestyle is gradually transformed

Table 11 The scale and efficiency effect.

Dep. Var: Electricity consumption	Efficiency effect				Scale effect			
	Urban		Rural		Urban		Rural	
	Diff-GMM	Sys-GMM	Diff-GMM	Sys-GMM	Diff-GMM	Sys-GMM	Diff-GMM	Sys-GMM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
L.Electricity	0.687*** (0.054)	0.814*** (0.029)	0.858*** (0.030)	0.969*** (0.018)	0.693*** (0.055)	0.813*** (0.027)	0.874*** (0.028)	0.978*** (0.021)
Urbanization	0.106*** (0.040)	0.066*** (0.011)	0.144* (0.079)	-0.017 (0.031)	0.085 (0.060)	0.126*** (0.025)	0.070** (0.092)	-0.032 (0.059)
<i>Urbanization * Energyintensity</i>	-0.047 (0.036)	-0.055** (0.026)	0.091** (0.036)	0.071*** (0.022)				
<i>Urbanization * Industrial</i>					0.021 (0.142)	-0.171*** (0.062)	0.172** (0.355)	0.047 (0.083)
Other control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constants		1.009*** (0.149)		0.221*** (0.078)		1.077*** (0.141)		0.111 (0.102)
Sargan	0.983	0.964	0.521	0.924	0.978	0.950	0.478	0.901
A-B test for AR (1)	0.010	0.016	0.000	0.000	0.010	0.016	0.000	0.000
A-B test for AR (2)	0.181	0.203	0.901	0.734	0.183	0.203	0.822	0.693
<i>N</i>	540	570	540	570	540	570	540	570

Note: *Urbanization * Energyintensity* is the interaction term between urbanization and energy intensity; *Urbanization*Industrial* is the interaction term between urbanization and industrial rate; Standard errors in parentheses; ***, ** and * represent significance at the 1%, 5% and 10% levels, respectively.

into an “industrial mode”, which means work takes up more of their time¹⁷ and thus workers’ daily household electricity consumption is reduced.

As discussed above, the scale and efficiency effect change over urban and rural areas. We believe that the scale effect, especially the scale effect of rural areas, is dominant before the turning point. Thus, urban-rural electricity consumption inequality is decreasing at this stage. After the turning point, the negative efficiency effect of urban areas causes electricity consumption drop off. However, at the same time, rural areas’ positive efficiency effect constantly increases electricity consumption. Eventually, the rural electricity consumption of some provinces exceeds the urban. That interaction of scale and efficiency effect is probably a mechanism for the U-shaped relationship between urbanization and urban-rural electricity consumption inequality.

7 Conclusions

In this paper, we have studied the impact of urbanization on urban-rural electricity consumption inequality. To answer this question, we modify a standard STRIPAT model and decompose the inequality determinants into urbanization, urban-rural income disparity and other socio-economic components. By using a provincial panel data including 30 China’s provinces over the period of 1997 to 2016, we carry out the empirical estimation mainly based on the first difference GMM and system GMM model. Following Halkos and Paizanos’s (2013) method, we further estimate the long-run effects of each independent variable on the basis of estimated short-run coefficients.

¹⁷The statistic of the Organization for Economic Co-operation and Development (OCED) reports that Chinese workers’ yearly working time is about 2,000 to 2,200 hours, American is about 1790 hours, Dutch is about 1419 hours, German is about 1371 hours and Japan is about 1719 hours.

We find that, generally, there is a robust long-run U-shaped relationship between the urbanization and urban-rural electricity consumption inequality. The estimated turning point arrives at the urbanization level of around 60%. The U-shaped relationship remains when we take several alternative specifications of regressors and sample divisions. Besides, Our estimates also show that both the increasing urban-rural income gap and the urban-rural education gap will exacerbate urban-rural electricity consumption inequality. However, the inequality is negatively associated with the infrastructure construction funds and the rise of fuel price.

To further explore the impacts of sample regions' geographical and economic differences on our baseline results, we perform a regional heterogeneity analysis. We then have two interesting findings: firstly, the colder northern region's turning point arrives later than the south, specifically, northern region's turning point is around the urbanization rate of 71.48%-74.98% and the south is around 55.38%-59.61%. Secondly, the baseline U-shaped relationship between urbanization and urban-rural electricity consumption inequality is valid for developed eastern regions, while for the undeveloped midwest, the relationship is not nonlinear but negative linearly.

By empirically investigate the mechanism underlying such U-shaped relationship, we explicate urbanization's impact on inequality through the scale effect and efficiency effect. We find that the scale effect, especially the scale effect of rural areas, is dominant before the turning point. After the turning point, the negative efficiency effect of urban areas causes electricity consumption drop off. However, meanwhile, rural areas' positive efficiency effect constantly increases electricity consumption. Eventually, this interaction of scale and efficiency effect determines the U-shaped relationship between urbanization and urban-rural electricity consumption inequality.

These results remind China's policymakers that there is no one universal policy for both urban and rural energy development, and energy policies should adapt to local urbanization. Specifically, for developed eastern regions, improving power efficiency and ameliorating energy behavior of rural areas ought to be the emphasis of local energy policies. However, it is still urgent for relatively undeveloped midwestern provinces to constantly promote urbanization and improve rural areas' electricity availability. A point of future research would be to design an optimal policy to balance urban and rural's electricity consumption.

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